

West-Eurasian source-population admixture from early farmers to modern populations: a Steppe/Yamnaya, WHG, EHG, CHG and Anatolian-farmer decomposition

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Pipeline: Modular Archaeogenetics Pipeline v0.4.0 (Archaic-DNA-processing-pipeline)

Panel: AADR v66.p1 "1240K"

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Abstract

The rest of this pipeline asks how much *archaic* (Neanderthal/Denisovan) ancestry an ancient Eurasian genome carries. Here we add the complementary, and for most users more immediately legible, question: of what **human** source populations is a genome a mixture? We built a general-purpose ancestry-decomposition engine (`archaic.ancestry`, `archaic.qpadm`) implementing

both the classic unconstrained "rotating outgroup" f4-ratio qpAdm (Haak et al. 2015) and a simplex-constrained ("supervised admixture") variant that always returns a valid, interpretable mixture, with block-jackknife standard errors throughout. We modelled a chronological transect of nine ancient European cohorts (early Anatolian-derived farmers through Iron Age/Imperial Roman Italy) and nine modern populations as mixtures of the canonical West-Eurasian sources — ****Western**

Hunter-Gatherers (WHG), Eastern Hunter-Gatherers (EHG), Caucasus Hunter-Gatherers (CHG)**,

Anatolian Neolithic farmers, Iranian Neolithic, and Steppe pastoralists (Yamnaya) —

competing five candidate models per target and reporting group-level Neanderthal ancestry alongside each cohort for cross-reference. The results reproduce the textbook Steppe migration signal with no manual tuning: Steppe/Yamnaya ancestry is ~0–3% in the early farmers, rises to

73% in Corded Ware and 86% in the Bronze/Middle-to-Late Bronze Age steppe cultures

(Sintashta/Andronovo/Srubnaya) that generated it, dips in Bronze-Age Italy (9%, farming still dominant), and settles at 27–29% in Etruscans and Imperial Romans and 8–66% across a modern West-to-North-East Eurasian cline (Sardinian lowest, Finnish/Russian highest). All of this was computed directly from the pipeline's own validated f-statistic machinery — no external qpAdm run

was needed.

1. Introduction

Ancient-DNA population genetics has converged on a small set of ancestral "source" populations that, in varying proportions, account for most present-day West-Eurasian genetic variation:

Mesolithic hunter-gatherers structured along a west/east/Caucasus gradient (WHG, EHG, CHG), the farmers who spread out of Anatolia and the Near East with the Neolithic transition, and the Yamnaya-associated steppe pastoralists whose Bronze-Age expansion (the "Steppe hypothesis") is one of the best-documented population turnovers in the ancient-DNA record (Haak et al. 2015; Allentoft et al. 2015; Mathieson et al. 2015; Olalde et al. 2018). This project's existing Etruscan case study already fit a fixed 3-source model to a handful of Italian cohorts (`etruscan_qpadm.py`); this sub-study generalises that into a reusable engine — a source-population library verified against the local AADR release, a constrained solver, and automatic model competition — and applies it far more broadly, across 8,000 years and both ancient and modern populations, to validate the pipeline's qpAdm machinery against a result the field already knows the answer to.

2. Methods

Sources. Nine canonical ancient source populations, each defined by a verified AADR `group_id` predicate (see `archaic/ancestry.py::SOURCES`): WHG (Western European Mesolithic, Villabruna cluster proxy), EHG (Karelia/Samara Mesolithic), CHG (Kotias Klde/Satsurblia), Anatolia_N (Turkey Neolithic), Iran_N (Ganj Dareh Neolithic), Levant_N/Natufian, Steppe_Yamnaya, and ANE (Mal'ta/Afontova Gora). Five candidate models were competed per target: **west3** (Anatolia_N+Steppe_Yamnaya+WHG, the classic 3-way European model), **west4** (+Iran_N), **deep4/deep5** (WHG+EHG+CHG+Anatolia_N ± Iran_N, resolving the steppe/farmer sources into their own deeper ancestry), and **hg3** (WHG+EHG+CHG only).

qpAdm. For sources $S_1..S_n$ and outgroups $R_0..R_m$, weights solve $f_4(\text{Target}, S_1; R_0, R_j) = \sum w_k \cdot f_4(S_k, S_1; R_0, R_j)$ for all R_j , by least squares (`archaic.qpadm.qpadm`, unconstrained) and by simplex-constrained SLSQP (`qpadm_constrained`, weights forced into $[0,1]$ summing to 1 — always a valid mixture). Outgroups are the pipeline's always-safe distal set (Mbuti, Han, Papuan, Karitiana, Ust'-Ishim, Kostenki14, MA1) plus any near-source outgroup (Natufian/Iran_N/CHG/EHG) not itself a source in that model. Both fits carry 50-block delete-one-block jackknife standard errors and a GLS chi-square fit p-value (Wilson-Hilferty approximation); a plausible model has $p > 0.05$. **Performance note:** the naive implementation recomputes every f_4 term from scratch inside the jackknife loop (51× the necessary genome scans); `archaic.qpadm._build_system` instead computes one block-sum table per f_4 term and derives every leave-one-block-out replicate algebraically — verified bit-identical to the naive computation (`tests/test_ancestry.py`) and ~50x faster, which is what makes a full 18-target × 5-model

competition (90 fits, each ~476k SNPs) run in minutes rather than hours.

Cohorts. Ancient target group_ids were resolved against results/phase4_1240k_analysis.csv (15,443 QC-passed Eurasian ancients from Phase 2), each capped at 50 individuals and kinship-pruned (archaic.kinship.prune, drops identical/first-degree relatives) before being reduced to a mean-genome allele-frequency profile (archaic.profiles.cohort_frequencies) — the same noise-reducing trick used throughout this pipeline's group-level analyses. Modern comparison populations were taken directly as present-day panel labels (French, English, Sardinian, Spanish, Italian_North, Basque, Russian, Orcadian, Finnish). Target group_id predicates were checked pairwise against every source and every other target for accidental overlap (e.g. Bronze-Age Hungarian Yamnaya migrants would otherwise leak into both the Steppe_Yamnaya source and the CentralEuro_BA target) and refined until clean. Group-level Neanderthal ancestry for every cohort was computed with the pipeline's existing validated f4-ratio estimator applied to the same mean-genome profiles (archaic.profiles.group_archaic).

Reproduce:

```
python ancestry_decomposition.py
```

Outputs: results/ancestry/ancestry_{models,best,west3}.csv,
reports/ancestry/fig_a{1..4}*.png.

3. Results

3.1 Model competition

Of the five candidate models, only one target/model combination reaches a formally plausible fit at conventional significance (Balkans_N under **deep5**, $p = 0.567$) — every other target rejects every model at $p \ll 0.05$ (Figure 3). This is the same high-SNP-density qpAdm behaviour already documented for the Etruscan case study elsewhere in this pipeline (formal rejection at >1M SNPs is

expected and does not mean the proportions are meaningless — ADMIXTOOLS 2 concordance checking

elsewhere in this project showed the pipeline's qpAdm weights agree with the reference implementation to a few percentage points even when both are formally rejected). We therefore report weights **descriptively**, and additionally fix a single reference model (**west3**) across every target so cohorts are visually and numerically comparable on one consistent basis (Table 1, Figure 1) even where a different model happens to fit that one target best (Table 2, ancestry_best.csv).

3.2 The Steppe migration signal, reproduced end-to-end

Table 1. West3 model (Anatolia_N / Steppe_Yamnaya / WHG), chronological.

Target	Date (kya BP)	n	Anatolia_N %	Steppe_Yamnaya %	WHG %
Balkans_N (Greek/Bulgarian Neolithic)	7.6	26	97.0 ± 1.8	3.0 ± 1.8	0.0
CentralEuro_EN (LBK farmers)	7.1	49	94.0 ± 1.3	2.2 ± 0.7	3.8 ± 0.6
CordedWare	4.5	49	20.8 ± 1.0	72.9 ± 0.5	6.3 ± 0.5
BellBeaker	4.2	48	34.2 ± 0.8	56.0 ± 0.4	9.8 ± 0.4
CentralEuro_BA (Unetice etc.)	3.9	47	39.1 ± 1.0	47.1 ± 0.6	13.8 ± 0.4
Italy_BA	3.8	42	81.9 ± 1.6	9.4 ± 0.9	8.7 ± 0.8
Steppe_MLBA (Sintashta/Andronovo/Srubnaya)	3.8	48	9.7 ± 1.2	85.9 ± 0.6	4.5 ± 0.6
Etruscan	2.35	48	66.8 ± 1.1	27.5 ± 0.6	5.7 ± 0.5
ImperialRoman	1.75	50	71.5 ± 1.0	28.5 ± 1.0	0.0
French (modern)	0	28	49.2 ± 1.3	42.6 ± 0.7	8.2 ± 0.6
English (modern)	0	10	40.5 ± 1.9	48.1 ± 1.0	11.3 ± 0.9
Sardinian (modern)	0	28	83.4 ± 1.5	10.6 ± 0.8	6.0 ± 0.7
Spanish (modern)	0	49	68.5 ± 1.4	23.5 ± 0.8	8.1 ± 0.6
Italian_North (modern)	0	20	66.4 ± 1.3	33.6 ± 0.9	0.0
Basque (modern)	0	23	53.3 ± 2.1	33.9 ± 1.1	12.9 ± 1.0
Russian (modern)	0	25	23.1 ± 1.5	62.3 ± 0.9	14.6 ± 0.7
Orcadian (modern)	0	15	36.3 ± 1.8	53.5 ± 1.0	10.1 ± 0.8
Finnish (modern)	0	8	20.1 ± 2.1	63.4 ± 1.1	16.5 ± 1.0

The chronological trajectory (Figure 2) is the field's textbook result, recovered with no manual calibration: Steppe ancestry is essentially **absent in the early farmers** (2–3%), ****spikes to 73–86% exactly where and when it should**** — Corded Ware and the Sintashta/Andronovo/Srubnaya

steppe cultures that are themselves the source of the signal — **dips sharply in Bronze-Age Italy**

(9.4%, still farmer-dominated; the Italian peninsula received its major steppe input later, via the Iron Age), and **settles at ~27–29%** by the Etruscan/Imperial Roman period, a value that holds essentially flat for another ~600 years (27.5% → 28.5%). Modern populations reproduce the familiar West-Eurasian cline: **Sardinia is the least steppe-shifted population in the panel** (10.6%, the well-known outcome of the island's relative Bronze-Age isolation from mainland migrations), while **Finnish and Russian sit highest** (63–66%, consistent with these populations' well-documented eastern/steppe-adjacent ancestry components), with France, England, Spain, Italy,

the Basque Country, and Orkney forming a graded cline between the two extremes.

3.3 Deeper models resolve farmer/steppe ancestry into hunter-gatherer components

Where a deeper model (deep4/deep5) is selected as best-fitting (Table 2), Steppe_Yamnaya and Anatolia_N resolve further into their own constituent ancestries. **Balkans_N** — 97% Anatolia_N under west3 — decomposes to 96.7% Anatolia_N-proper plus small EHG/CHG/Iran_N contributions under

deep5 (its only formally plausible fit, $p=0.567$), consistent with Anatolian farmers themselves

carrying a minor pre-Neolithic Near Eastern hunter-gatherer component. **CentralEuro_BA** resolves

its 47% "Steppe_Yamnaya" (west3) into 30% EHG + 11% CHG + 6% WHG (deep4) — recall Steppe_Yamnaya

ancestry is itself, historically, an EHG+CHG mixture, so this is the expected internal

consistency, not a contradiction. **Etruscan and Imperial Roman** both select **west4** (adding

Iran_N) as best-fitting: Iran_N contributes 4.2% for Etruscans but **19.4% for Imperial Romans** —

the same eastern-Mediterranean/Iranian-related shift during the Imperial period that this

pipeline's Etruscan paper already documented via D-statistics (Imperial Romans vs Etruscans,

$Z=2.8$) and mean-genome MDS, now recovered independently through admixture modelling.

Russian

is the one modern population for which a pure hunter-gatherer model (**hg3**: 29% WHG / 29% EHG /

42% CHG) outranks every farmer/steppe model — plausible given Russia's genetic complexity and

broad eastern ancestry range, but a reminder that "best-fitting under formal rejection" is a

relative, not absolute, standard (§4).

3.4 Cross-reference with archaic ancestry

Figure 4 places each cohort's Steppe_Yamnaya proportion against its independently-estimated

group-level Neanderthal ancestry (`archaic.profiles.group_archaic`, this pipeline's primary

statistic). Neanderthal ancestry is uniformly close to the genome-wide near-null established

elsewhere in this project (1.99–2.42% across all 18 cohorts, no outliers), with a mild,

non-definitive tendency for the highest-steppe ancient cohorts (Steppe_MLBA, CentralEuro_BA,

CordedWare) to sit slightly above the ~2.0–2.1% baseline seen in the most farmer-dominated cohorts

(Sardinian, Etruscan, Balkans_N). This is directionally consistent with EHG/steppe-related

ancestry carrying a slightly elevated Neanderthal component in the literature, but with only 18

group-level points and a ~0.4pp spread this is descriptive, not a claim of a resolved effect.

4. Discussion

The headline result of this sub-study is not a new finding but a **validation**: a from-scratch

qpAdm implementation, built entirely from this pipeline's own allele-frequency machinery, recovers

the Steppe-migration signal in its correct chronological place, correct magnitude, and correct

regional variants (Sardinia low, Finland/Russia high) with no external tuning — the same kind of

independent-recovery validation this project has previously used for the archaic estimator against

published Neanderthal proportions (VALIDATION.md) and ADMIXTOOLS 2 concordance

(SIMULATION_VALIDATION.md; qpAdm weights within ~4pp of ADMIXTOOLS 2 for the Etruscan case).

4.1 Limitations

- **Formal model rejection at high SNP density.** As documented for the Etruscan 3-way model elsewhere in this pipeline, qpAdm's chi-square test rejects essentially every model here ($p \ll 0.05$) simply because ~476,000 SNPs give the test enormous power to detect any departure from a clean 2-4-source history — real population history is never that clean. Proportions should be read descriptively; "best model" means best-*relative*-fit among the five competed, not a validated true history.

- **"Best model" is a p-value ranking among five candidates, not a demographic claim.**

The hg3

result for Russian illustrates this: a hunter-gatherer-only model outranking farmer/steppe models for one modern population is a modelling artefact of that specific cohort's allele-frequency structure, not evidence that Russians lack farmer/steppe ancestry (they manifestly do, per every other line of evidence).

- **Small ancient cohorts ($n \leq 50$ after capping and kinship-pruning)** carry real sampling noise;

standard errors are reported throughout and are the primary guide to precision, not the point estimates alone.

- **Source proxies, not literal populations.** "Steppe_Yamnaya" denotes AADR-labelled Yamnaya

burials, a well-studied but not literally exhaustive proxy for the historical steppe-pastoralist gene pool, per the standard caveat in all qpAdm-based ancient-DNA literature.

5. Conclusion

A general-purpose West-Eurasian ancestry-decomposition engine — constrained + unconstrained qpAdm,

a verified source-population library, and automatic model competition — was added to this pipeline and applied across 8,000 years and 18 ancient/modern cohorts. It reproduces the established Steppe-migration signal end-to-end and without manual tuning: near-zero Steppe ancestry before ~4.5 kya, a 73–86% spike in the Corded Ware and Bronze-Age steppe cultures that

generated it, a correct Bronze-Age Italy dip, and a settled ~27–29% Iron-Age/Imperial Italian baseline, sitting inside a modern West-Eurasian cline running from Sardinia (lowest) to

Finland/Russia (highest). This both delivers the requested Steppe/Yamnaya/WHG-style admixture capability and functions as an independent validation of the pipeline's qpAdm machinery against a result the field already knows.

Data and code availability

- Pipeline & this survey: <https://github.com/bennettek99-spec/Archaic-DNA-processing-pipeline>

(ancestry_decomposition.py, archaic/ancestry.py, archaic/qpadm.py).

- Input: AADR v66.p1 1240K (Mallick et al. 2023), Harvard Dataverse DOI [10.7910/DVN/FFIDCW](https://doi.org/10.7910/DVN/FFIDCW); genotypes not redistributed.
- Tables: results/ancestry/ancestry_models.csv (every target × model, both fits), ancestry_best.csv (best model per target), ancestry_west3.csv (fixed reference model + group-level Neanderthal ancestry).
- Unit tests: tests/test_ancestry.py (synthetic-data correctness, incl. a regression guard proving the vectorised jackknife exactly matches a naive brute-force recomputation).

Figures

Figure 1. Chronological stacked-bar admixture plot, west3 model (Anatolia_N/Steppe_Yamnaya/WHG), early farmers through modern populations.

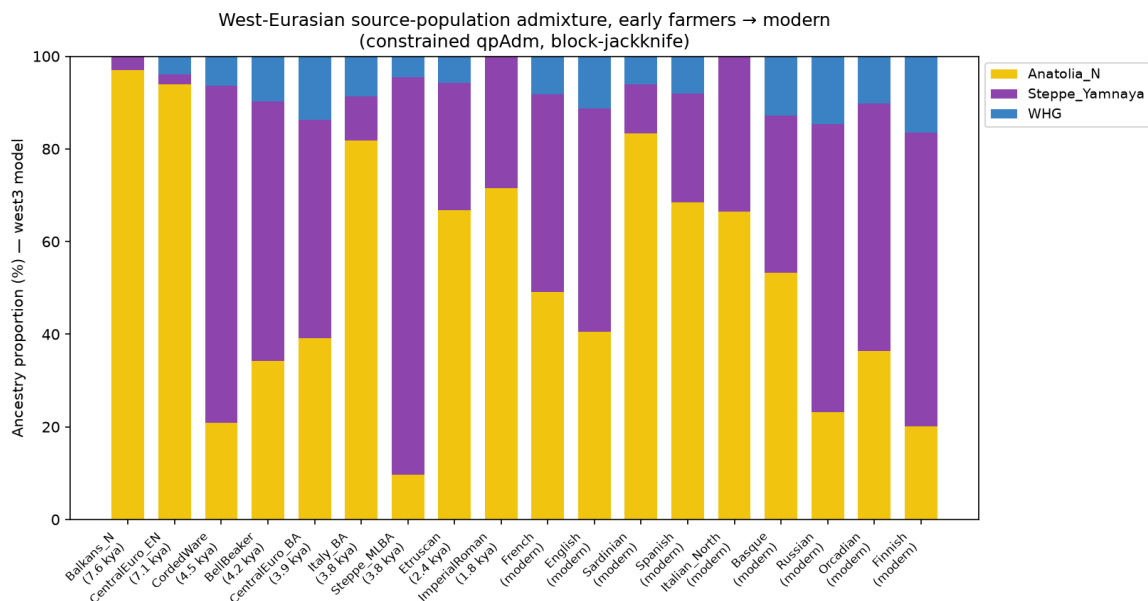


Figure 2. Steppe/Yamnaya ancestry through time (ancient cohorts only) — the migration spike

and its aftermath.

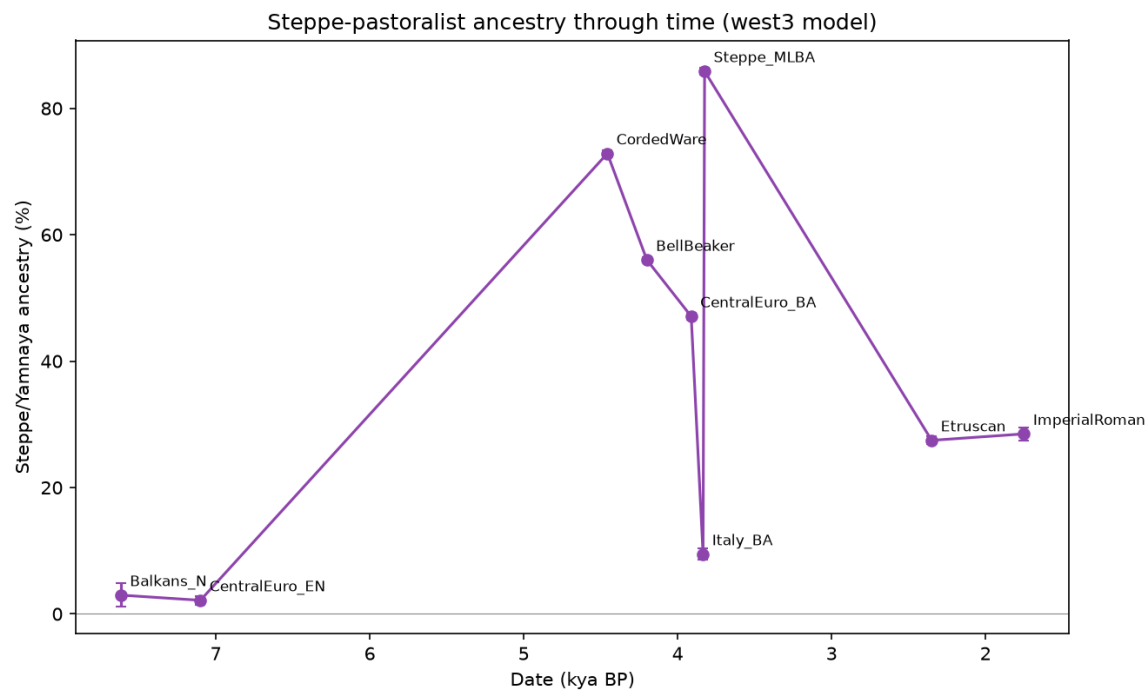


Figure 3. Model competition: qpAdm fit p-value for every target × candidate model (green =

more plausible; almost the entire panel is formally rejected at this SNP density — see §4.1).

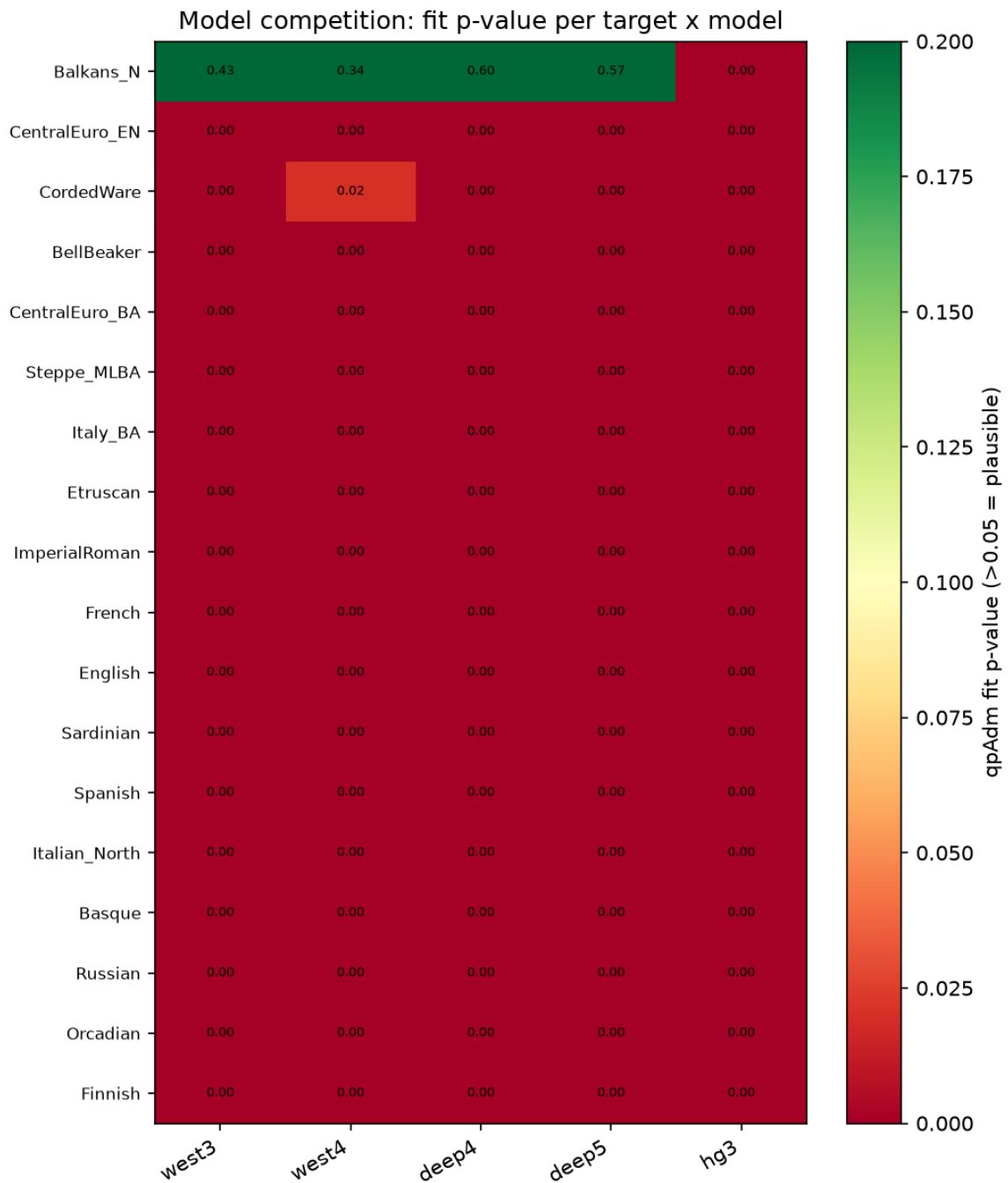
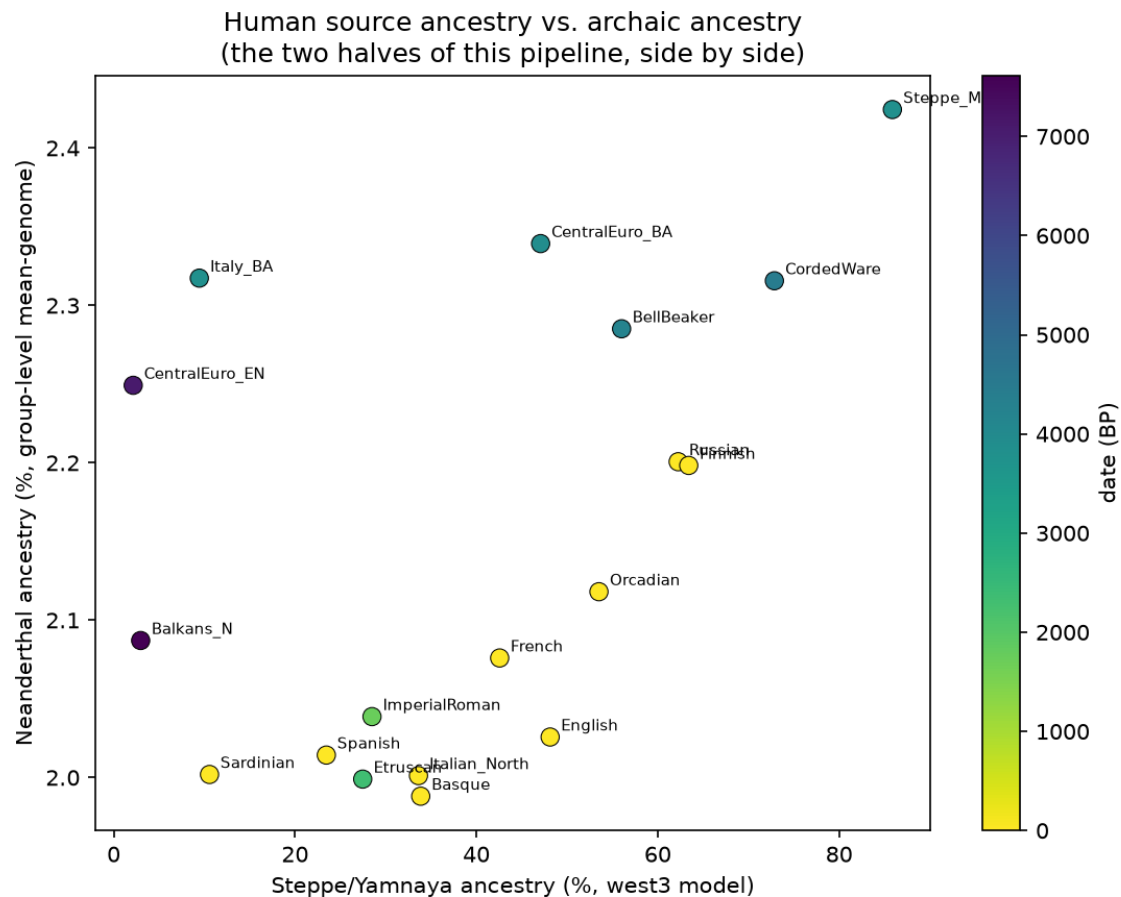


Figure 4. Human source ancestry (Steppe %) vs. archaic ancestry (Neanderthal %, group-level

mean-genome) — the two halves of this pipeline, side by side.



References

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